



2021 International Solid-State Circuits Conference Tutorial



T9: Designing Amplifiers for Stability

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Most amplifier designers spend at least as much or more effort assuring stability under all operating conditions with proper frequency compensation than on other amplifier features. This tutorial will revisit the basics of frequency compensation such as Miller, parallel, nested-Miller and feed-forward compensation and compare them in terms of power efficiency and load-drive capabilities. We will look into the loading effects of subsequent stages, common design traps and round up with a case study of recently published amplifiers.

Viola Schaffer was born in Szeged, Hungary in 1974. She received the M.S. degree in electrical engineering from the University of Arizona, Tucson in 1999.

She joined Texas Instruments Incorporated (formerly Burr-Brown Corporation) in 1998 and has been working as an analog IC design engineer/manager at various locations including Tucson, Arizona, as well as Erlangen and Freising in Germany. She was elected Distinguished Member Technical Staff in 2018. Her work focuses on precision signal conditioning including instrumentation and programmable-gain amplifiers, power amplifiers, industrial drivers as well as magnetic-based current sensors. She holds 17 patents related to this work with several applications pending.